Lecture 10 v2:
An Introduction to Panel Data

25117 - Econometrics

Universitat Pompeu Fabra

November 20th, 2024

What we learned in the last lesson

- Instrumental variables regression is a way to estimate causal coefficients when one or more regressors are correlated with the error term.
- Endogenous variables are correlated with the error term in the equation of interest; exogenous variables are uncorrelated with this error term.
- For an instrument to be valid, it must be (1) correlated with the included endogenous variable and (2) exogenous.
- IV regression requires at least as many instruments as included endogenous variables.
- The TSLS estimator has two stages.
  - First, the included endogenous variables are regressed against the included exogenous variables and the instruments.
  - Second, the dependent variable is regressed against the included exogenous variables and the predicted values of the included endogenous variables from the first-stage regression(s).
- Weak instruments (instruments that are nearly uncorrelated with the included endogenous variables) make the TSLS estimator biased and TSLS confidence intervals and hypothesis tests unreliable.
- If an instrument is not exogenous, the TSLS estimator is inconsistent.

Starting with Pooled Data

Cross-Section

So far we have been mostly focused on **cross-sectional data** — a type of data collected at a single point in time or over a very short period, capturing information from multiple subjects or entities at that specific moment.

In other words, it provides a snapshot or a "cross-section" of a population or a phenomenon at a particular point in time.

A Cross-Sectional Data Set on Wages and Other Individual Characteristics

obsno	wage	educ	exper	female	married
1	3.10	11	2	1	0
2	3.24	12	22	1	1
3	3.00	11	2	0	0
4	6.00	8	44	0	1
5	5.30	12	7	0	1
⋮	⋮	⋮	⋮	⋮	⋮
525	11.56	16	5	0	1
526	3.50	14	5	1	0

Pooled Cross-Section

A natural extension of the cross-sectional data is the **pooled cross-sectional data** — where cross-sectional data is gathered at multiple points in time and where each cross-sectional study is independent of the others.

In other words, different samples of subjects or entities are observed during each data collection period.

Pooled Cross Sections: Two Years of Housing Prices

	obsno	year	hprice	proptax	sqrft	bdrms	bthrms
Year 1993	1	1993	85500	42	1600	3	2.0
	2	1993	67300	36	1440	3	2.5
	3	1993	134000	38	2000	4	2.5
	⋮	⋮	⋮	⋮	⋮	⋮	⋮
	250	1993	243600	41	2600	4	3.0
Year 1995	251	1995	65000	16	1250	2	1.0
	252	1995	182400	20	2200	4	2.0
	253	1995	97500	15	1540	3	2.0
	⋮	⋮	⋮	⋮	⋮	⋮	⋮
	520	1995	57200	16	1100	2	1.5

Pooled Cross-Section

Many surveys, randomly sampling individuals, families, or firms are repeated at regular time intervals (e.g., CPS in the U.S.)

Pooled together $\Longrightarrow$ independently pooled cross section

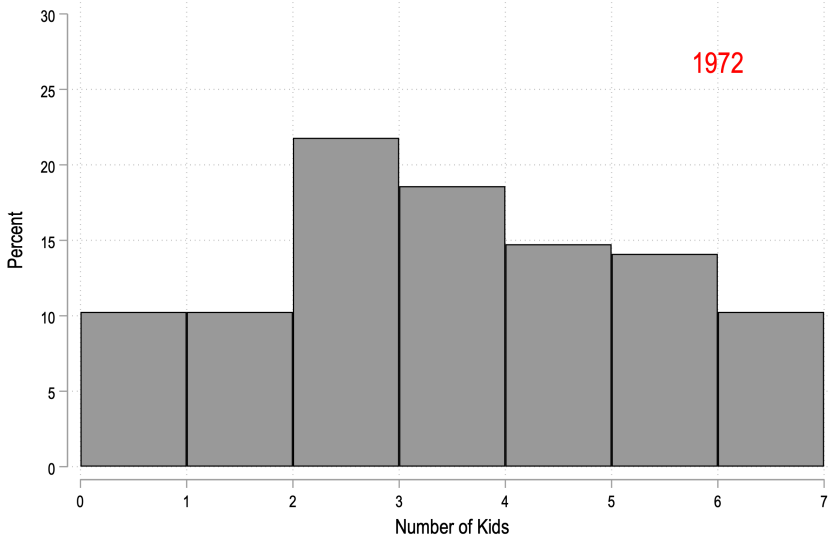
Pros

- Statistical inference is the same as for cross-sectional methods
- Larger sample size (i.e., lower standard errors)
- Enables us to estimate trends conditional on explanatory variables.
- Enables us to estimate how the effect of one factor has changed over time (e.g., how did the gender wage gap change?)

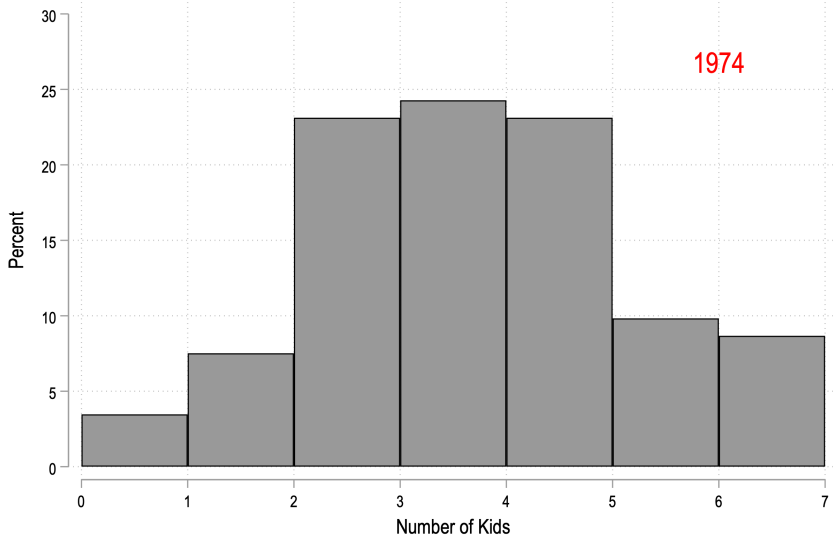
Cons

- Populations have different distribution across time periods, thus they are not *identically* distributed
  $\rightarrow$ we can allow the intercept to differ across periods! *(how?)*

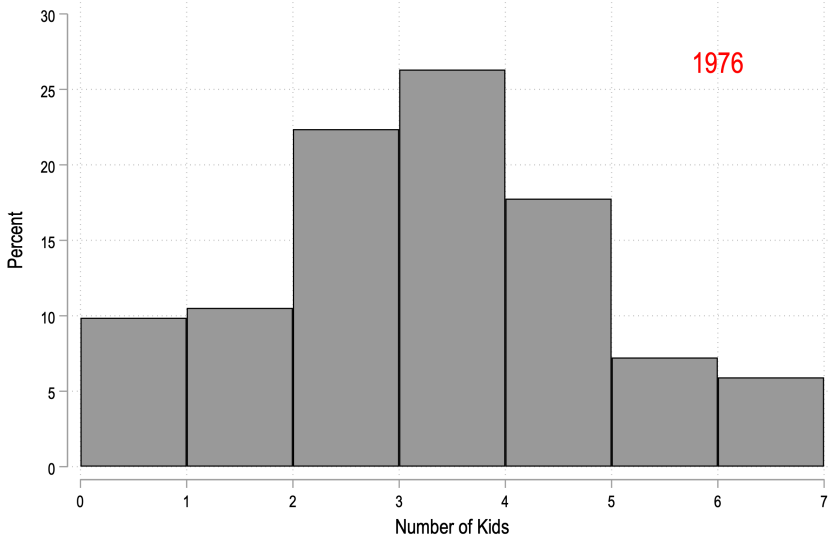
Fertility over time



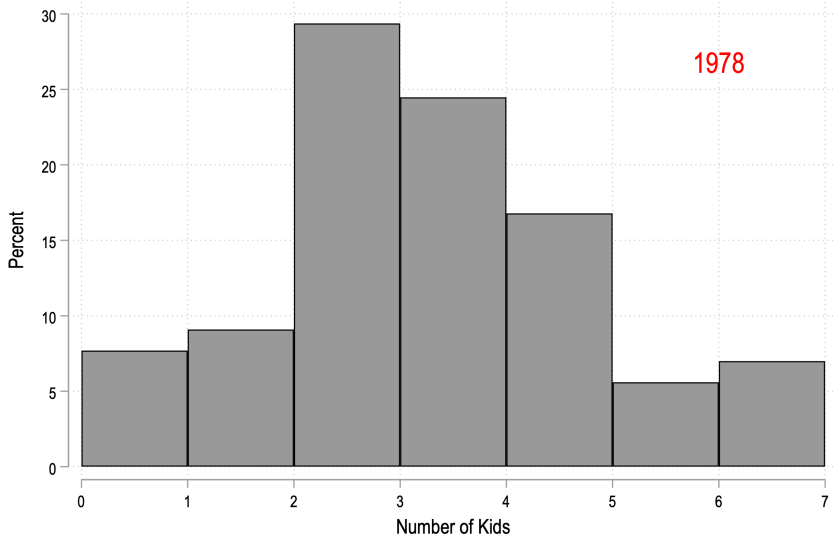
Fertility over time



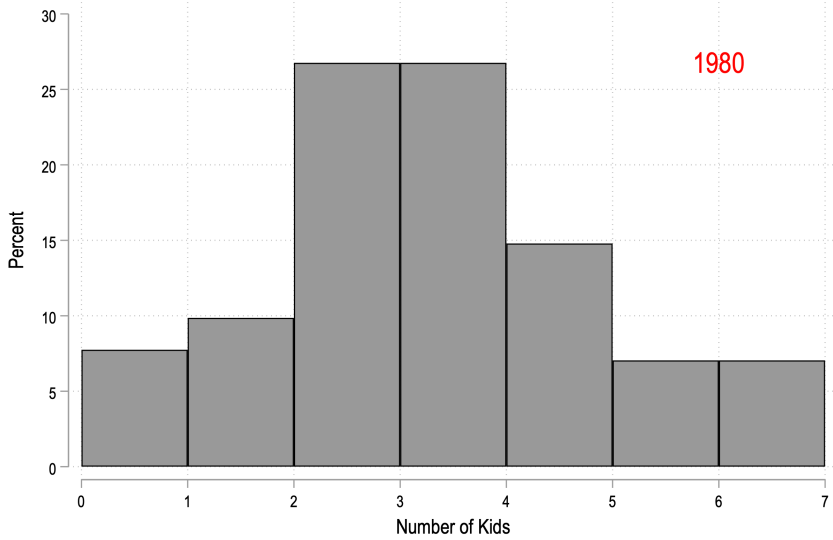
Fertility over time



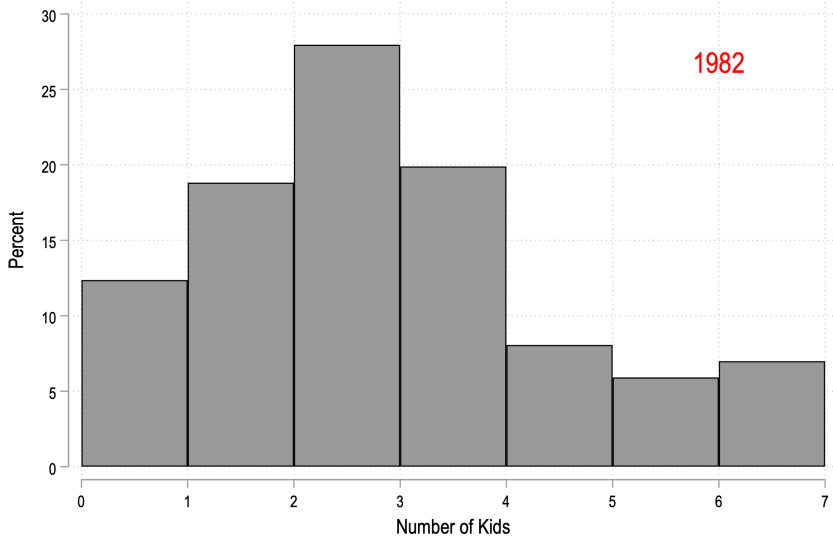
Fertility over time



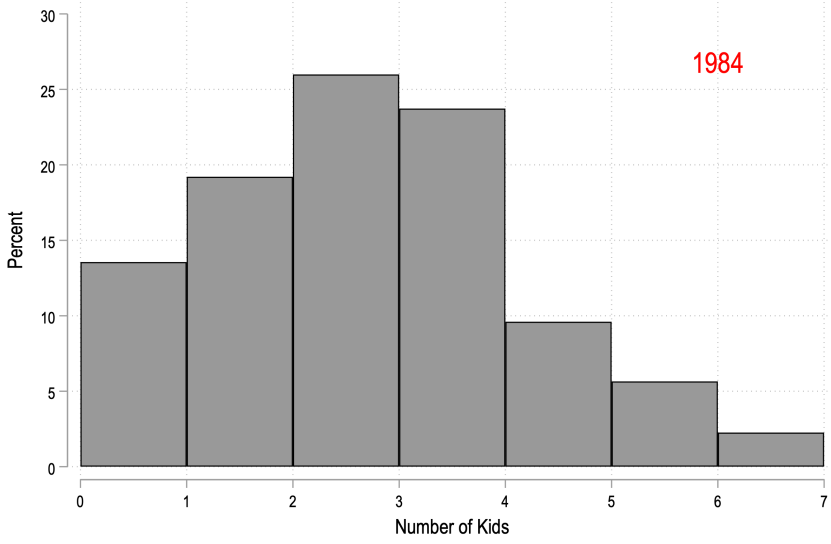
Fertility over time



Fertility over time



Fertility over time



Fertility over time

From Lecture 7 — We can use year-specific intercepts by including year-specific dummies

Remember from Lecture 7 — the base year here is...?

Sharp decline in the number of kids per woman in the 1980s which is *not* explained by changes in main demographics

Selected year coefficients

Year (base: 1972)	Coefficient
1974	0.2682
1976	−0.0974
1978	−0.0687
1980	−0.0713
1982	−0.5225
1984	−0.5452

$n = 1{,}129,\ R^2 = 0.1295,\ SER = 1.5548.$

Controls: education, age, age squared, race, region, and type of residence.

Gender Gap

As in Lecture 7, we can allow for different intercepts *and* slopes gradients by interacting a dummy and a continuous variable

Example:

- Pooled 2 cross-sectional datasets (1978 and 1985)
- Data on wages, educational attainment, and gender
- *Questions:*
  - Did the gender wage gap change from 1978 to 1985?
  - Did the return to education change from 1978 to 1985?

$$\begin{aligned} ln(wage) &= \beta_0 + \beta_1 \mathbb{1}_{1985} + \beta_2 Educ + \beta_3 (Educ \times \mathbb{1}_{1985}) + \beta_4 Female + \beta_5 (Female \times \mathbb{1}_{1985}) \\ &\quad + \beta_6 Experience + \beta_7 Experience^2 + \beta_8 Union + u \end{aligned}$$

Gender Gap

Dependent variable: log wage

Regressor	Coefficient
Education	0.0747209
1985 × Education	0.0184605
Female	−0.3167086
1985 × Female	0.0850520
Experience	0.0295843
Experience squared	−0.0003994
Union	0.2021319
1985	0.1178062
Intercept	0.4589329

$n = 1{,}084,\ R^2 = 0.4262,\ SER = 0.4127.$

Testing Differences in Pooled regressions

When *all* variables are interacted with the time dummies, the regression specification is equivalent to estimating separate regressions — one for each year.

As in Lecture 7, one can test whether the population regression lines differ across two different years by running a simple **Chow test**.

First, run the fully interacted model ($year_1$ serves as the base period)

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki} + \beta_{k+1} \mathbb{1}_{year_2} + \beta_{k+2}(\mathbb{1}_{year_2} \times X_{1i}) + \cdots + \beta_{2k+1}(\mathbb{1}_{year_2} \times X_{ki}) + u_i$$

Then test $\beta_{k+1} = \beta_{k+2} = \cdots = \beta_{2k+1} = 0$

We can also test whether the intercepts differ between the 2 groups ($H_0 : \beta_{k+1} = 0$) or whether the slopes differ ($H_0 : \beta_{k+2} = \cdots = \beta_{2k+1} = 0$)

From Pooled to Panel Data

What is a panel dataset?

A panel dataset (also called longitudinal data) contains observations on multiple entities (e.g., individuals, states, companies, etc.), observed at two or more points in time (e.g., days, months, years, decades, etc.).

For example,

- Data on 420 California school districts in 1999 and again in 2000, for 840 observations total.
- Data on 50 U.S. states, each state is observed in 3 years, for a total of 150 observations.
- Data on 1000 individuals in four different months, for 4000 observations total.

Panel data with k regressors: $\{X_{1,it}, X_{2,it}, ..., X_{k,it}, Y_{it}\}$ where $i = 1, \ldots, n$ and $t = 1, \ldots, T$

When all $k + 1$ variables are observed for all units i at all points in time t (i.e., no missing observations), we say the panel is balanced.

Panel Data

Panel data is also known as **longitudinal** data or **cross-sectional time-series** data

It involves observations of multiple entities over multiple time periods.

We say a panel is **balanced** if all entities are observed in each time period, and there are no missing observations.

We will focus on the case where $N >> T$

Same city

Different years

TABLE 1.5

A Two-Year Panel Data Set on City Crime Statistics

obsno	city	year	murders	population	unem	police
1	1	1986	5	350000	8.7	440
2	1	1990	8	359200	7.2	471
3	2	1986	2	64300	5.4	75
4	2	1990	1	65100	5.5	75
⋮	⋮	⋮	⋮	⋮	⋮	⋮
297	149	1986	10	260700	9.6	286
298	149	1990	6	245000	9.8	334
299	150	1986	25	543000	4.3	520
300	150	1990	32	546200	5.2	493

Why are panel data useful?

With panel data, we can control for factors that:

- Vary across units but not over time (in the sample period) ⟵ (we will focus on that case first)
- Vary over time (in the sample period) but not across units

These 'black boxes' could contain omitted variables, which are typically unobserved or unmeasured — and therefore cannot be included in the regression using multiple regression.

For example,

- Average individual ability or preferences
- Cultural attitudes
- First geography (distances, topography, etc.) components
- Aggregate trends

Here's the key idea:

- If an omitted variable does not change over time or across units, then any changes in Y over time and across units cannot be caused by the omitted variable.

Example of a panel data set

You want to study the impact of alcohol taxes on traffic death.

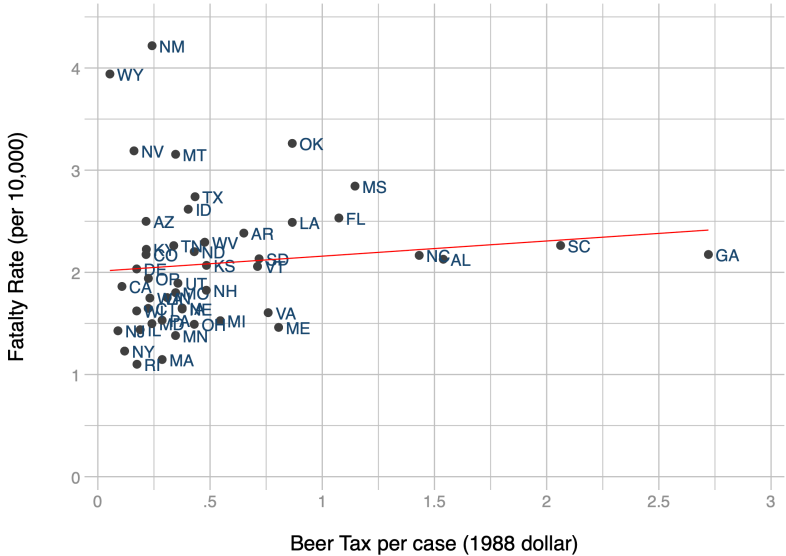
- Observational unit: U.S. state ($n = 48$), yearly ($T = 7$) between 1982 and 1988
- Balanced panel, so total observations $= 7 \times 48 = 336$
- Variables:
  - Traffic fatality rate (# traffic deaths in that state in that year, per 10,000 state residents)
  - Tax on a case of beer
  - Other (legal driving age, drunk driving laws, etc.)

Consider the simple following specification:

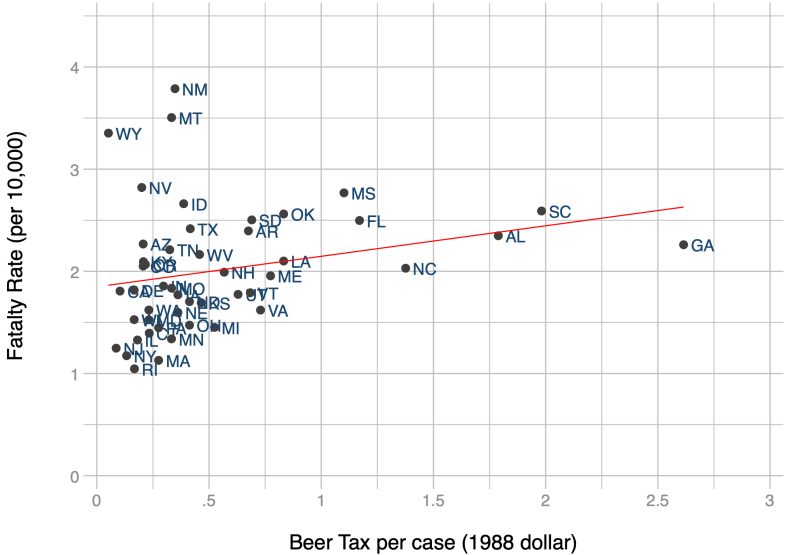
$$Fatalities_{i,t} = \beta_0 + \beta_1 BeerTax_{i,t} + u_{i,t}$$

where t is a specific year, so we are just looking at individual cross-sections...

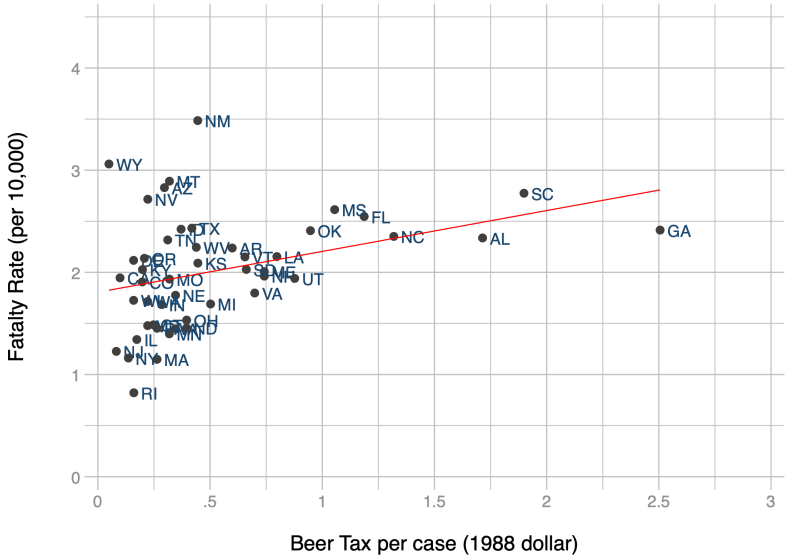
Traffic Fatalities vs. Beer Tax — 1982



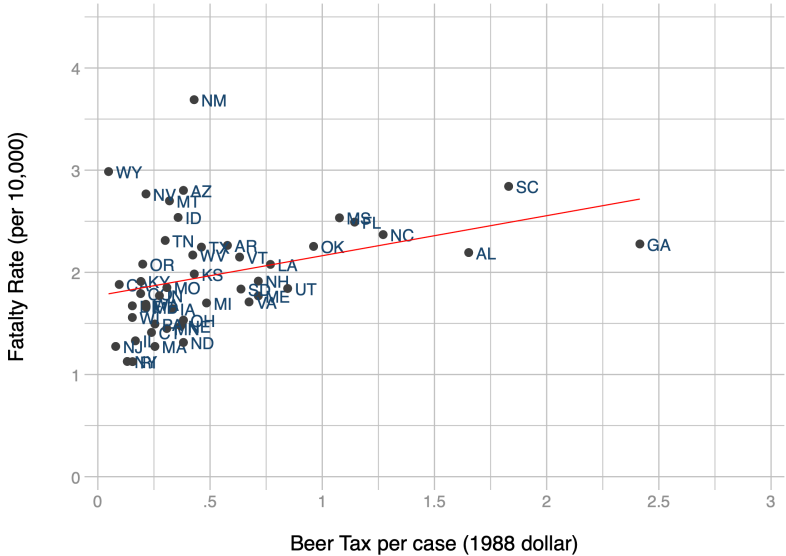
Traffic Fatalities vs. Beer Tax — 1983



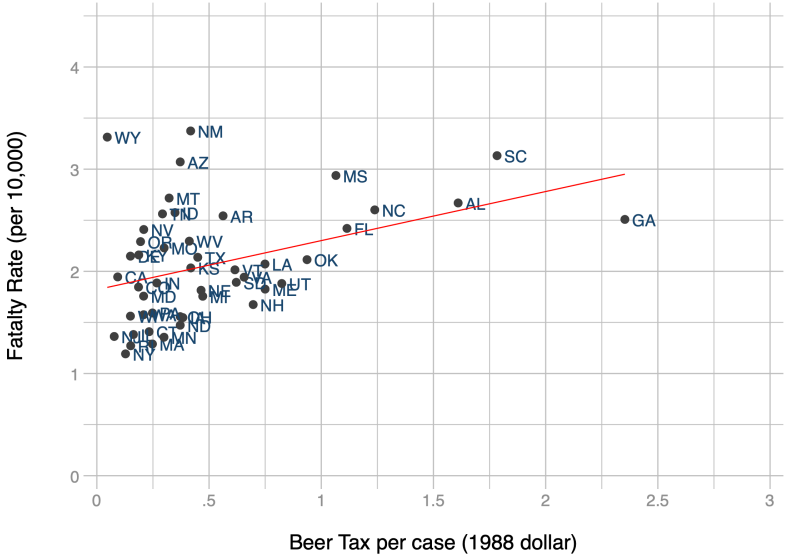
Traffic Fatalities vs. Beer Tax — 1984



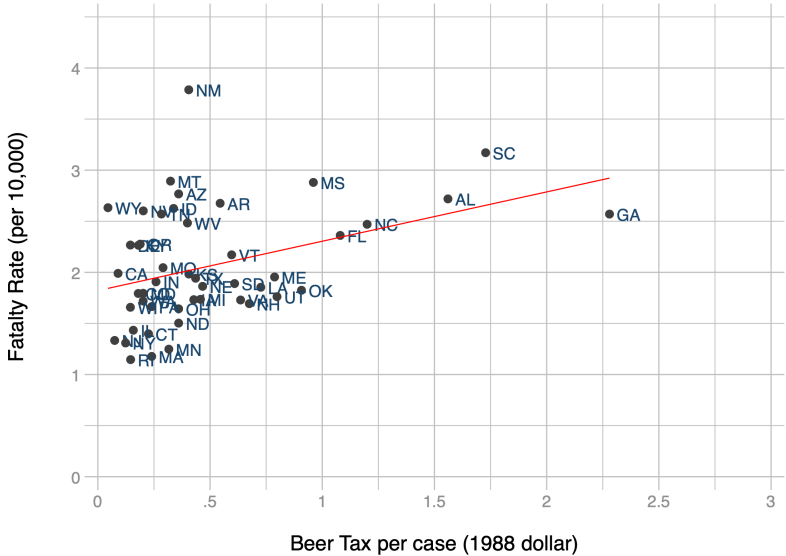
Traffic Fatalities vs. Beer Tax — 1985



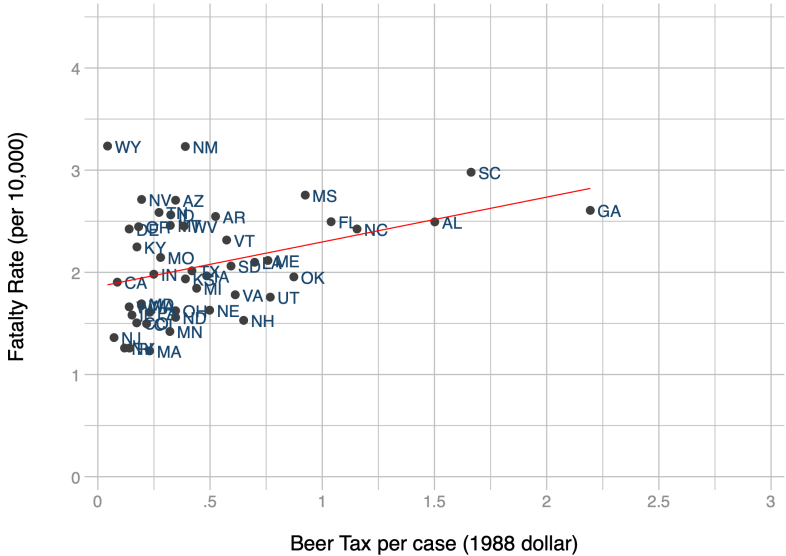
Traffic Fatalities vs. Beer Tax — 1986



Traffic Fatalities vs. Beer Tax — 1987



Traffic Fatalities vs. Beer Tax — 1988



Traffic Fatalities vs. Beer Tax – First Differences

Simple analysis of individual cross sections return weird results... Higher alcohol taxes, more traffic deaths? Arguably, omitted factors could cause omitted variable bias (which ones?)

Now consider the simple panel specification:

$$Fatalities_{i,t} = \beta_0 + \delta_0 \mathbb{1}_{1988} + \beta_1 BeerTax_{i,t} + \beta_2 Z_i + u_{i,t}$$

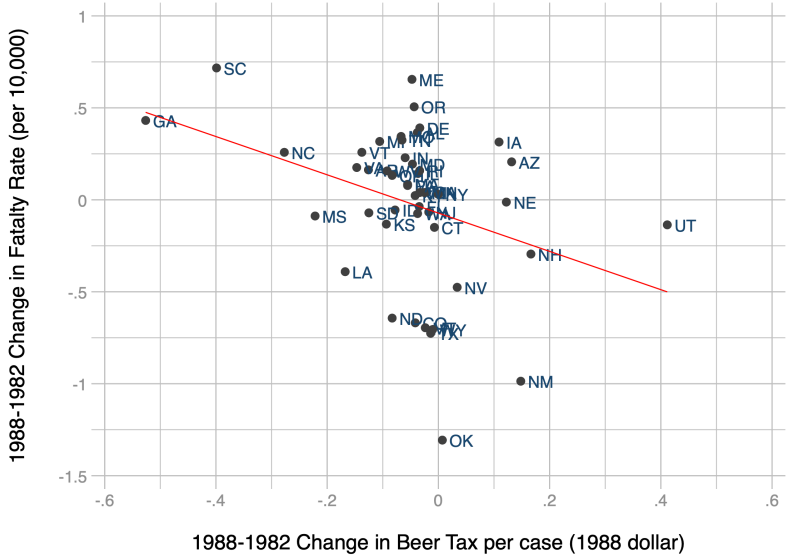
where $t \in \{1982, 1988\}$.

If Z_i is not observed, we could run in an OVB issue. But taking the first difference eliminates the value of Z_i!

$$Fatalities_{i,1988} - Fatalities_{i,1982} = \delta_0 + \beta_1 (BeerTax_{i,1988} - BeerTax_{i,1982}) + (u_{i,1988} - u_{i,1982})$$

This "difference" equation can be estimated by OLS, even though Z_i is unobserved!

Traffic Fatalities vs. Beer Tax – First Differences



First Differences when $T > 2$

We can also use differencing with more than two time periods:

$$Fatalities_{i,t} = \beta_0 + \delta_1 \mathbb{1}_{1983} + \delta_2 \mathbb{1}_{1984} + \cdots + \delta_6 \mathbb{1}_{1988} + \beta_1 BeerTax_{i,t} + \beta_2 Z_i + u_{i,t}$$

That is,

$$\begin{aligned}
Fatalities_{i,1982} &= \beta_0 + \beta_1 BeerTax_{i,1982} + \beta_2 Z_i + u_{i,1982} \\
Fatalities_{i,1983} &= \beta_0 + \delta_1 \mathbb{1}_{1983} + \beta_1 BeerTax_{i,1983} + \beta_2 Z_i + u_{i,1983} \\
Fatalities_{i,1984} &= \beta_0 + \delta_2 \mathbb{1}_{1984} + \beta_1 BeerTax_{i,1984} + \beta_2 Z_i + u_{i,1984} \\
&\cdots \\
Fatalities_{i,1988} &= \beta_0 + \delta_6 \mathbb{1}_{1988} + \beta_1 BeerTax_{i,1988} + \beta_2 Z_i + u_{i,1988}
\end{aligned}$$

We do so by taking differences between adjacent time periods:

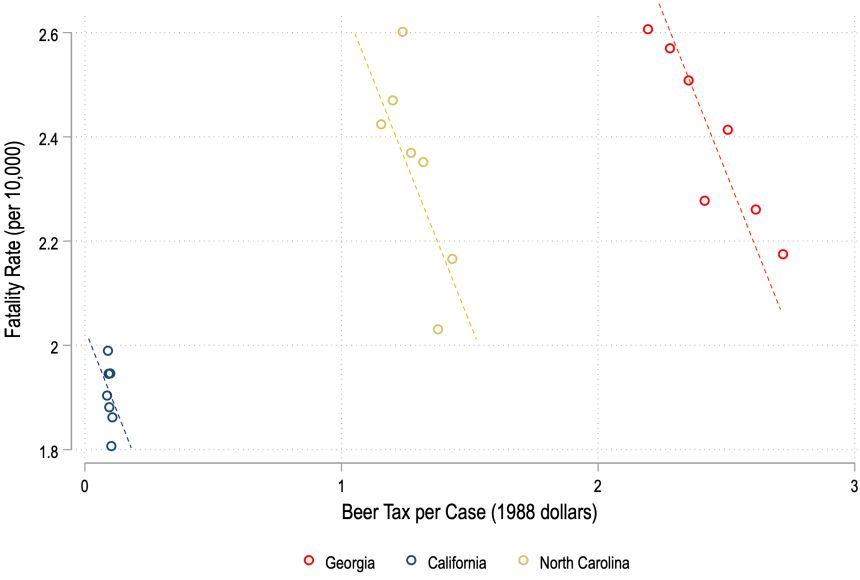
$$\Delta Fatalities_{i,t} = \delta_1 + \Delta\delta_2 \mathbb{1}_{1984} + \cdots + \Delta\delta_6 \mathbb{1}_{1988} + \beta_1 \Delta BeerTax_{i,t} + \Delta u_{i,t}$$

Limits of First-Differences

- The main limitation of using FD compared to other methods is probably that we lose the initial period (there are $N(T-1)$ observations)
- Because FD builds on ensuing observations, missing data can be very problematic (imagine if we only observe some i once every other year)
- Typically, the variation in the differenced independent variable is much smaller than the variation in the original independent variable. Thus, imprecise estimates could happen from the FD estimator.
- Serial correlation (more on that later)
- First differencing can enlarge classical errors-in-variable (i.e., measurement error) bias

Fixed Effects Models


Fatality Rate (per 10,000)
2.6
2.4
2.2
2
1.8
0
1
2
3
Beer Tax per Case (1988 dollars)
Georgia
California
North Carolina


Fatality Rate (per 10,000)
2.6
2.4
2.2
2
1.8
0
1
2
3
Beer Tax per Case (1988 dollars)
Georgia
California
North Carolina

Fixed Effects Regressions

The general FE model reads

$$Fatalities_{i,t} = \beta_0 + \beta_1 BeerTax_{i,t} + \beta_2 Z_i + u_{i,t}$$

For example, in the case of California,

$$\begin{aligned} Fatalities_{CA,t} &= \beta_0 + \beta_1 BeerTax_{CA,t} + \beta_2 Z_{CA} + u_{CA,t} \\ &= (\beta_0 + \beta_2 Z_{CA}) + \beta_1 BeerTax_{CA,t} + u_{CA,t} \\ &= \alpha_{CA} + \beta_1 BeerTax_{CA,t} + u_{CA,t} \end{aligned}$$

Where α_{CA} is the intercept for California, and β_1 is the slope.

The intercept is unique to CA, but the slope is the same in all the states: parallel lines.

Fixed Effects Regressions

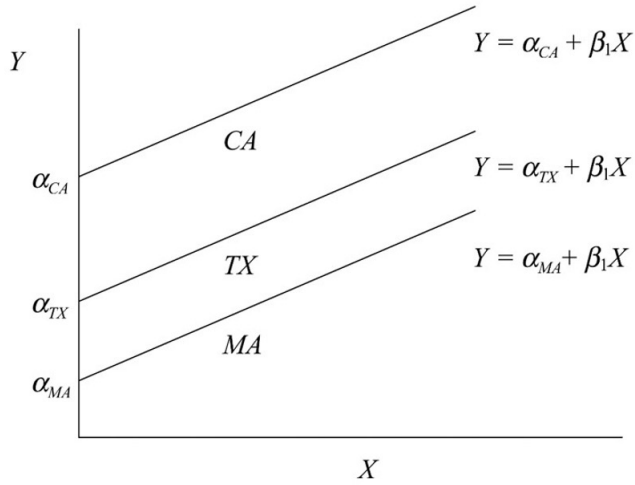
Two equivalent ways of writing the same specification:

$$Y_{i,t} = \beta_0 + \beta_1 X_{i,t} + \gamma_1 D_i \mid_{i=1} + \gamma_2 D_i \mid_{i=2} + \cdots + \gamma_n D_i \mid_{i=n} + u_{i,t}$$

where $D_i \mid_{i=n} = 1$ if $i = n$ (i.e., state $\#n$).

And,

$$Y_{i,t} = \alpha_i + \beta_1 X_{i,t} + u_{i,t}$$



Fixed Effects Regressions

Including n extra (fixed effect) dummies can be computationally intensive when n is large. What regression softwares typically do is to (time-)*demean* (**within transformation**) the specification:

$$\bar{Y}_i = \alpha_i + \frac{1}{T}\sum_{t=1}^{T} \beta_1 X_{i,t} + \frac{1}{T}\sum_{t=1}^{T} u_{i,t}$$

Which yields to

$$Y_{i,t} - \bar{Y}_i = \beta_1 (X_{i,t} - \frac{1}{T}\sum_{t=1}^{T} X_{i,t}) + (u_{i,t} - \frac{1}{T}\sum_{t=1}^{T} u_{i,t})$$

which can be estimated by simple OLS, that is

$$\tilde{Y}_{i,t} = \beta_1 \tilde{X}_{i,t} + \tilde{u}_{i,t}$$

and β_1 isolates the impact of changes in X within a unit, ignoring the constant, unit-specific factors influencing both X and Y.

Fixed Effects Regressions

State fixed effects: traffic fatalities and beer tax

Regressor	Coefficient	Clustered SE
Beer tax	−0.6558736	0.2918556
Reported intercept	2.3770750	0.1497966

336 observations, 48 states, 7 years per state; within $R^2 = 0.0407$.

Standard errors are clustered by state to allow serial correlation within a state.

```
xtreg vfrall beertax, fe vce(cluster state)
```

The `fe` option includes state fixed effects. Individual state effects are omitted from this table; the reported intercept depends on their normalization.

Fixed Effects Regressions

By analogy, we can also include time FEs (i.e., they are the same across unit but vary in time). What omitted variable could be captured by time FEs in the traffic fatalities vs. beer taxes case?

The population line for, say, 1985 reads

$$\begin{aligned} Fatalities_{i,1985} &= \beta_0 + \beta_1 BeerTax_{i,1985} + \beta_2 Z_{1985} + u_{i,1985} \\ &= (\beta_0 + \beta_2 Z_{1985}) + \beta_1 BeerTax_{i,1985} + u_{i,1985} \\ &= \lambda_{1985} + \beta_1 BeerTax_{i,1985} + u_{i,1985} \end{aligned}$$

Where λ_{1985} is the intercept for 1985, and β_1 is the slope.

The dummies notation holds, as in the preceding example with unit FEs.

Fixed Effects Regressions

Including T extra (fixed effect) dummies can be computationally intensive when T is large. As before, we can transform the panel model specification into a simple OLS one by demeaning:

$$\bar{Y}_t = \lambda_t + \frac{1}{n}\sum_{i=1}^{n} \beta_1 X_{i,t} + \frac{1}{n}\sum_{i=1}^{n} u_{i,t}$$

Which yields to

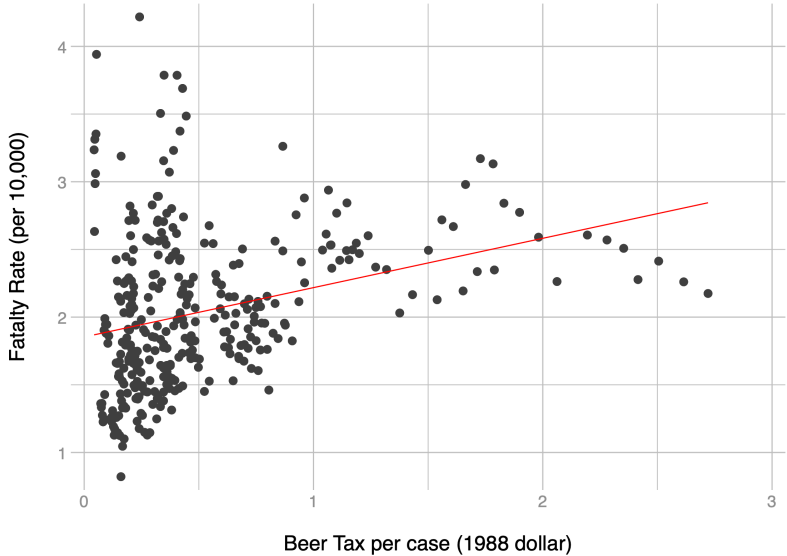
$$Y_{i,t} - \bar{Y}_t = \beta_1 (X_{i,t} - \frac{1}{n}\sum_{i=1}^{n} X_{i,t}) + (u_{i,t} - \frac{1}{n}\sum_{i=1}^{n} u_{i,t})$$

which can be estimated by simple OLS, that is

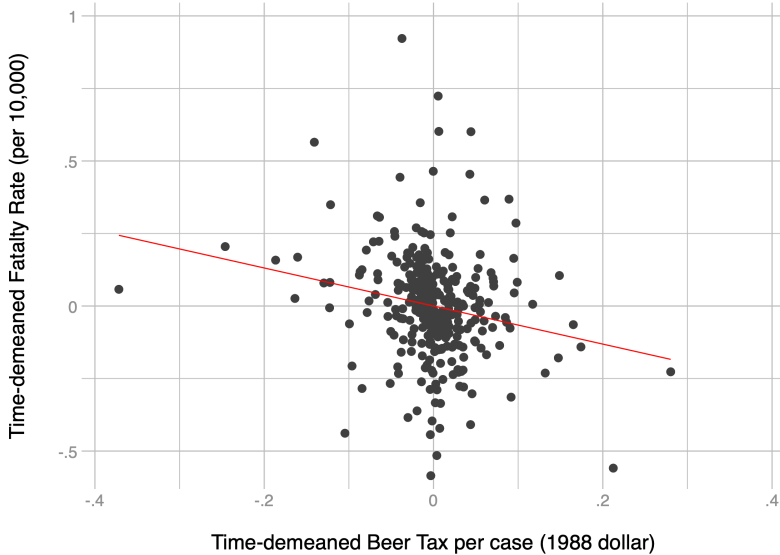
$$\check{Y}_{i,t} = \beta_1 \check{X}_{i,t} + \check{u}_{i,t}$$

and β_1 captures the effect of changes in X within a specific time period, accounting for shared temporal factors that affect both X and Y at the same time.

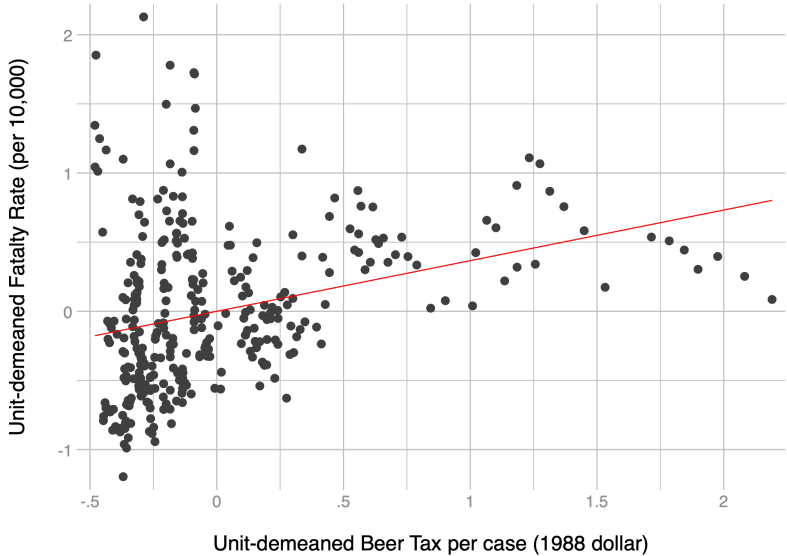
Raw Panel Data



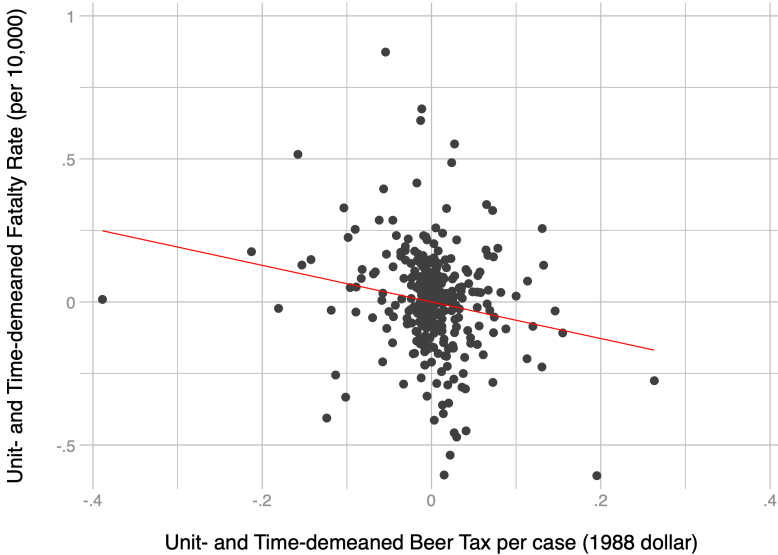
Accounting for α_i



Accounting for λ_t



Both unit and time fixed effects



Both unit and time fixed effects

State and year fixed effects

Regressor	Coefficient	Clustered SE
Beer tax	−0.6399799	0.3570783
1983	−0.0799029	0.0350861
1984	−0.0724206	0.0438809
1985	−0.1239763	0.0460559
1986	−0.0378645	0.0570604
1987	−0.0509021	0.0636084
1988	−0.0518038	0.0644023
Reported intercept	2.4284700	0.2016885

336 observations, 48 states; within $R^2 = 0.0803$. Base year: 1982.

Standard errors are clustered by state; state effects are omitted from the table.

The Fixed Effects Simple Regression Assumptions

Consider the following regression model

$$Y_{it} = \alpha_i + \beta_1 X_{i,t} + u_{i,t}$$

1. $\mathbb{E}(u_i \mid X_{i1}, X_{i2}, \ldots, X_{iT}, \alpha_i, \lambda_t) = 0$
2. $(X_{i1}, X_{i2}, \ldots, X_{iT}, u_{i1}, u_{i2}, \ldots, u_{iT})$ are i.i.d. draws from their joint distribution
3. Large outliers are unlikely: $(X_{it}, u_{i,t})$ have fourth moments.
4. There is no perfect multicollinearity (multiple X's)

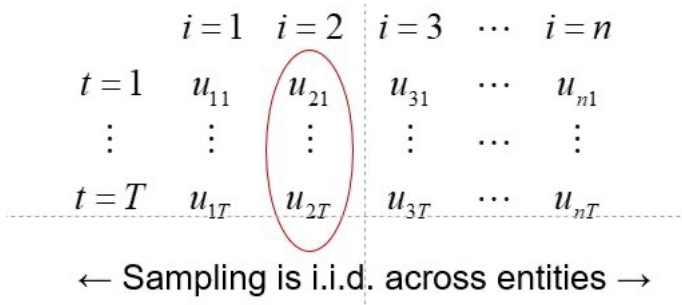
Assumption 1 means that the error term cannot be correlated with any present, past, or future value of X (no omitted lagged effects, no feedback loop!)

Assumption 2 means that variables are independent across units but makes no such restriction within a unit.

Assumption 3 and 4 did not change (see Lecture 7).

Serial Correlation

- Following up on Assumption 2, when $u_{i,t}$ is correlated over time within a unit, i.e., when $cov(u_{i,t}, u_{i,t+j}) \neq 0$ we say that $u_{i,t}$ is **autocorrelated**, or **serially correlated**.
- If entities are sampled by simple random sampling, then $(u_{i1}, u_{i2}, \ldots, u_{iT})$ is independent of $(u_{j1}, u_{j2}, \ldots, u_{jT})$...



- But in many panel data applications, $u_{i,t}$ is serially correlated

$\longrightarrow$ Clustering/Moulton's problem (1986)

Clustered SEs

The usual OLS standard errors will in general be wrong when u_{it} is serially correlated, because they omit covariances between errors at different dates.

In practice, they often understate the true sampling uncertainty: if u_{it} are correlated over time, you don't have as much information (as much random variation) as you would if u_{it} were uncorrelated!

To see this, consider the panel the panel (time-)demeaned regression estimator

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}\sum_{t=1}^{T}(X_{it} - \bar{X}_i)(Y_{it} - \bar{Y}_i)}{\sum_{i=1}^{n}\sum_{t=1}^{T}(X_{it} - \bar{X}_i)^2} = \frac{\sum_{i=1}^{n}\sum_{t=1}^{T}\tilde{X}_{it}\tilde{Y}_{it}}{\sum_{i=1}^{n}\sum_{t=1}^{T}\tilde{X}_{it}^2} = \beta_1 + \frac{\frac{1}{nT}\sum_{i=1}^{n}\sum_{t=1}^{T}\tilde{X}_{it}\tilde{u}_{it}}{\frac{1}{nT}\sum_{i=1}^{n}\sum_{t=1}^{T}\tilde{X}_{it}^2}$$

This completely mirrors the basic OLS estimator equation!!!

Clustered SEs

Rearranging the terms,

$$\sqrt{nT}(\hat{\beta}_1 - \beta_1) = \frac{\sqrt{\frac{1}{n}}\sum_{i=1}^{n}\sqrt{\frac{1}{T}}\sum_{t=1}^{T}\tilde{X}_{it}\tilde{u}_{it}}{\frac{1}{nT}\sum_{i=1}^{n}\sum_{t=1}^{T}\tilde{X}_{it}^2} = \frac{\sqrt{\frac{1}{n}}\sum_{i=1}^{n}\eta_i}{\hat{Q}_{\tilde{X}}}$$

Because η_i is i.i.d across units (by Assumption 2), of mean 0 (by Assumption 1), with finite variance (by Assumption 3), then (same reasoning as in Lecture 3!)

$$\sqrt{nT}(\hat{\beta}_1 - \beta_1) = \frac{\sqrt{\frac{1}{n}}\sum_{i=1}^{n}\eta_i}{\hat{Q}_{\tilde{X}}} \sim \mathcal{N}\Big(0, \frac{\sigma_\eta^2}{Q_{\tilde{X}}^2}\Big) \qquad \text{as } n \text{ grows large}$$

Now, note that

$$\sigma_\eta^2 = var\left(\sqrt{\frac{1}{T}}\sum_{t=1}^{T}\tilde{X}_{it}\tilde{u}_{it}\right) = \frac{1}{T}var\Big(\sum_{t=1}^{T}\tilde{X}_{it}\tilde{u}_{it}\Big) = \frac{1}{T}var\Big[\tilde{X}_{i1}\tilde{u}_{i1} + \cdots + \tilde{X}_{iT}\tilde{u}_{iT}\Big]$$

Clustered SEs

Remember that $var(X + Y) = var(X) + var(Y) + 2cov(X, Y)$... therefore

$$\sigma_{\hat{\eta}}^2 = \frac{1}{T}\Big[var(\tilde{X}_{i1}\tilde{u}_{i1}) + \cdots + var(\tilde{X}_{iT}\tilde{u}_{iT}) + 2cov(\tilde{X}_{i1}\tilde{u}_{i1}, \tilde{X}_{i2}\tilde{u}_{i2}) + \cdots + 2cov(\tilde{X}_{i,T-1}\tilde{u}_{i,T-1}, \tilde{X}_{iT}\tilde{u}_{iT})\Big]$$

The variance formula for independent observations misses the autocovariances in the final part of this equation. If errors are serially correlated, these terms must be included when estimating sampling uncertainty.

Clustered SEs for panel data allow errors to be correlated over time within each unit. Inference uses independent units and a sufficiently large number of units.

The clustered standard error is

$$SE(\hat{\beta}_1) = \sqrt{\frac{1}{nT}\frac{s_{\hat{\eta}}^2}{\hat{Q}_{\tilde{X}}^2}}$$

Final Remarks on Clustered SEs

We can cluster SEs by unit of observation (e.g., serial correlation within states), but also across units (e.g., regions above states) if we suspect potential correlation in observations within these higher-level clusters.

However, asymptotic inference (as n grows large) supposes we have a large number of clusters...

This is not always the case. When there are too few clusters and serial correlation, standard errors might be biased!

In the Traffic Fatalities vs. Beer Tax example, is 48 (states) a number of clusters that is large enough? What should be done when there are too few clusters? This is out of the scope of this (introductory) lecture!

Material I

– **Textbooks:**

- Introduction to Econometrics, 4th Edition, Global Edition, by Stock and Watson – Chapters 10 - 13